

Selecting individual lines in the blocks indicates the clarity with which the scanner can capture images. If Photoshop detects a solid block instead of the constituent lines, it indicates that the scanner cannot capture the finer details.

Warranty and support: Here we took into account the warranty period for the scanners, the number of authorised service centres appointed by the companies and the number of cities in which authorised service centres are present. The information was provided by the respective companies. The weightage allotted to this category was 10 per cent.

Value for money: This is a factor of the performance and features offered by the scanner as compared to its price. We computed this value by dividing the sum of a scanner's scores in the performance and features tests by its price. The weightage allotted to this category was 20 per cent.

How they fared

We found a wide variation in the performance and features offered by the scanners in each category. It was also quite surprising to note that some of the budget scanners actually outperformed most mid range scanners, while the high-end scanners were in a class of their own. What's obvious from our test results is that you can find the right scanner for any need depending on your budget.

Features

We looked for standard features such as the maximum resolution and colour depth supported, time-saving features such as one-touch scanning support, the bundled accessories and software and the quality of documentation and manuals provided.

Resolution: We gave primacy to optical resolution over interpolated resolution, which is the actual resolution that a scan-

ner can support (this is a hardware limitation), but it can be interpolated to a much higher value using software. In the budget segment we found that all scanners supported at least a 600x1200 dpi resolution with the exception of the Agfa SnapScan 310P—it was the only scanner that offered a 300x600 dpi resolution. The BenQ and Mustek scanners had the best drivers—both supported a maximum interpolated resolution of 19200x19200 dpi.

Among the mid range scanners, the Umax scanners came out tops—the Astra 4400 and 5400 were the only scanners in this category to support an optical resolution of 1200x1200, but as observed in the budget category, the BenQ ScanPrisa 640BT and Mustek 1200ED had the highest interpolated resolution (19200x19200 dpi) in this category as well.

In the high-end segment, the HP ScanJet 5400C and 5470C supported a whopping 2400x2400 dpi optical resolution and infinite interpolated resolution. This was reflected in the scanned images, which were very crisp and realistic.

Interface: Almost every vendor in the budget segment has switched over to the USB interface since it provides a faster scanning speed and plug-and-play connectivity. The only exceptions were Agfa SnapScan 310P and Canon CanoScan N640P. Surprisingly no scanner supported the USB 2.0 interface.

The scenario was a little different for mid range scanners—four of seven scanners tested supported the USB interface (BenQ ScanPrisa 640BT, Microtek ScanMaker V6UPL, Umax Astra 4400 and Astra 5400). Agfa SnapScan 1212P and Mustek 1200ED were the only scanners that used the older and slower parallel interface, and Microtek ScanMaker X6EL was the only SCSI scanner in this segment.

In the high-end segment, the Epson Perfection 1650 and Umax Astra 4000U had only a USB interface. The two HP models also had a parallel interface, while Microtek ScanMaker X12USL had a SCSI interface along with USB. The only short-



Scanning a Scanner

A scanner is a device that captures and converts images and documents to a digital format. It does so by using an array of photosensitive cells that detect light reflected off or transmitted through the object being scanned. This array of receptors, referred to as a charge-coupled device (CCD), measures light intensity and converts it into an electrical charge.

An analogue to digital converter (ADC) is required to digitise the information. Each cell in the CCD array then creates one pixel and for each pixel a certain number of bits is stored. The more bits of information assigned to each pixel, the better the image quality.

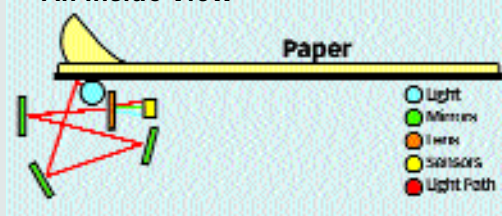
There are a few essential parts which go in the making of the scanner. These are:

CCD: The Charge Coupled Device is the heart of a scanner. It is made up of an array of photo diodes, which sense the light and

convert it into an electrical signal.

Lamp: This is usually a cold cathode fluorescent lamp. It lights up the area to be scanned with pure white light, which is

An Inside View



reflected back from the document and is captured by the CCD.

Scan head: This is the area which holds the CCD and the fluorescent lamp. The head assembly is mounted on the stabiliser bar, which moves from end-to-end while scanning.

Glass plate: This is where the document is placed. It is larger in size than the document (A3, A4) it supports. The scan quality depends a fair deal on the glass—if there is dirt over or under the glass, the scanned image will have artefacts and smudges.

Lens/mirror: The mirror reflects the light to the lens. There are usually two or three mirrors in a scanner. It is very critical for the placement of the mirror and the lens to be accurate so that all the light is concentrated on the CCD—otherwise some areas of the scanned image will be cut.

Stepper motor: The CCD, lamp, mirror and lens are all placed on the same head assembly. This assembly is moved by the stepper motor which takes its control signal from the circuitry placed at the bottom of the scanner.